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Toward real text manipulation detection: New dataset and new solution

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ABSTRACT

With the surge in realistic text tampering, detecting fraudulent text in images has gained prominence for maintaining information security. However, the high costs associated with professional text manipulation and annotation limit the availability of real-world datasets. With most relying on synthetic tampering, they inadequately replicate real-world tampering attributes. To address this issue, we present the Real Text Manipulation (RTM) dataset, encompassing 9,000 text images, which include 6,000 manually tampered images, created using a variety of techniques, alongside 3,000 unaltered text images for evaluating solution stability. Our evaluations indicate that existing methods falter in text forgery detection on the RTM dataset. We propose a robust baseline solution featuring a Consistency-aware Aggregation Hub and a Gated Cross Neighborhood-attention Fusion module for efficient multi-modal information fusion, supplemented by a Tampered-Authentic Contrastive Learning module during training, enriching feature representation distinction. This framework, extendable to other dual-stream architectures, demonstrated notable localization performance improvements of 3.99% and 5.76% on IoU and F1-measure, respectively. Our contributions aim to propel advancements in real-world text tampering detection. Code and dataset will be made available at https://github.com/DrLuo/RTM.

1. Introduction

Text manipulation detection (TMD) has recently attracted great interest owing to its importance in a wide range of applications such as digital finance [1], e-commerce [2], and certification audit [3]. Since text serves as a primary medium of information transmission, the sensitive contents carried by text are the usual targets of tampering for illegal purposes. With the rapid development of image editing tools [3,4], it becomes easier for tampered text images to deceive human eyes, which seriously threatens information security in daily life. Therefore, developing tampered text detection methods is urgently needed to defend against malicious tampering.

Actually, image manipulation detection (IMD), another hotspot in image forensics, has been studied more deeply with the increasing number of datasets [5–9]. The forged target is usually a conspicuous semantic object, such as the spliced two people in Fig. 1(a). In contrast, text manipulation usually aims to distort the carried contextual information. The tampered objects, *i.e.* texts, usually maintain similar visual semantics and appearances to the rest part of the document. The unique characteristics of text tampering bring in new challenges: (1) **Concealment**. Various factors lead to imperceptible visual appearance, *e.g.* small size, highly similar color, font, orientation and background. (2) **Fewer traces**. The environmental factors of text regions are usually simpler than semantic objects in natural scenarios. Some business documents are even simply composed of black text and white background. A simple background pattern would result in fewer manipulation traces (3) **Large variations**. Texts are made of characters, and the granularity of tampered text varies from characters to paragraphs. The manipulation techniques are diverse as well. Thus, existing IMD datasets are not fully appropriate for TMD tasks.

Nevertheless, constructing a real-world-oriented text manipulation dataset is not trivial. High-quality manual text manipulation requires professional skills, and the cost of text manipulation and fine-grained annotation is expensive. Hence, most recent datasets [2,10,11] are built through synthetic approaches. However, manual text manipulations account for a large portion of actual applications, whose characteristics are hard to be completely imitated by machines. Consequently, challenges are rooted in the gap between the recent benchmarks and the real-world scenarios, which we summarized as follow: (1) **Limited scale**: most currently available benchmarks contain only a limited number of data samples. As in Table 1, T-SROIE only consists of 986 tampered images. (2) **Lack of diverse manipulation techniques**: most currently available datasets [2,10] are manipulated in a specific

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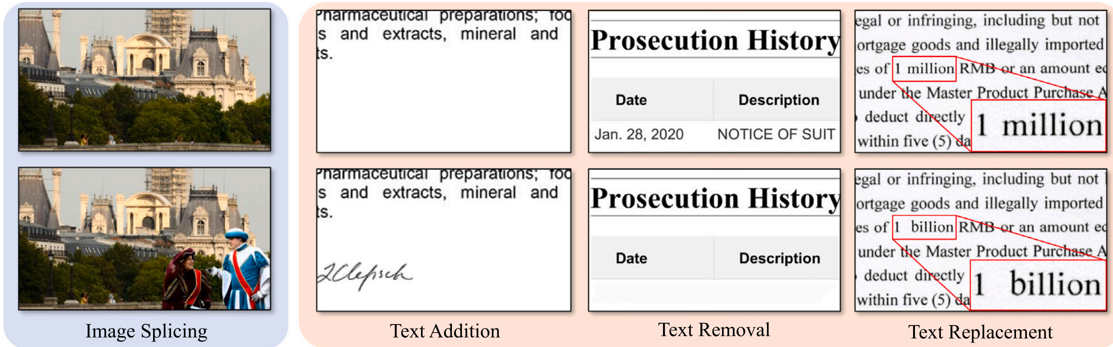


(a) Natural Image Manipulation (b) Text Manipulation

Fig. 1. (a) An example of splicing manipulation in a natural image. (b) Common types of tampered text images, including addition, removal and replacement. The red box zooms in the small region. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

manner, regardless of other possible as shown in Fig. 1(b). (3) **Lack of fidelity**: although recent generative techniques show promising ability to produce tampered text that can fraud human eyes, they are likely to leave inconsistent artifacts, such as blurry effects.

To bridge the gap, we introduce a new reality-oriented tampered text dataset named Real Text Manipulation (RTM), which consists of 9,000 images in total, including 6,000 tampered images manipulated through Adobe® Photoshop (PS) manually. Each image is annotated with a binary mask indicating the tampered pixels. We leverage PS layers to obtain consistent pixel-wise annotations for each image. Compared to the previous datasets, our dataset has several advantages. Firstly, RTM dataset leverages manual tampering conducted by professional editors, narrowing the gap with practical scenarios. Secondly, we adopt more types of manipulation techniques, including five basic manipulation operations and their combinations. Thirdly, the tampered regions in RTM have more diverse shapes and scales. Fourthly, we introduce authentic samples in our dataset to evaluate the image-level accuracy, which is also of great importance in practical applications.

With the proposed dataset, we build a new benchmark including both semantic segmentation and image manipulation detection methods. The experimental results demonstrate that current state-of-the-art algorithms fail to make satisfactory performance, especially against manual manipulations. The concealed nature of tampered texts increases the difficulty of detecting them. Furthermore, we build an extensible baseline called ASC-Former (Asymmetric Stream Contrastive Transformer) that achieves the best overall performance on RTM. In particular, an auxiliary branch is paralleled as an additional cue to trace manipulation artifacts, which can receive multiple transformed domains. A Consistency-aware Aggregation Hub (CA Hub) is designed to gather informative cues from each domain, which can be easily extended to more views with little computational expansion. Since the transformed domains are derived from the original RGB image and represent a certain local property of the input, we aggregate the multi-source features via a Gated Cross Neighborhood-attention Fusion module (GCNF) adaptively. In addition, a Tampered-Authentic Contrastive Learning module (TAC) is employed during the training phase to explicitly increase the discrimination of latent features, where a memory mechanism is equipped to enlarge sample pairs due to the imbalanced distribution of tampered pixels. Extensive Experiments show the effectiveness of the proposed designs.

Considering the diverse and reality-oriented characteristics owned by RTM, we hope that RTM could serve as a new benchmark to evaluate text manipulation detection methods in practice. To sum up, our contributions are summarized as follows:

1. We introduce a new dataset tailored for real-world text manipulation detection, which covers various manual manipulation techniques. It contains 9,000 images of 5 types of basic tampering operations that commonly occur in real-world scenarios. To the best of our knowledge, it is the largest text manipulation detection dataset toward handcraft operations with fine-grained annotations.
2. We analyze the characteristics of both the task and RTM dataset. We conduct a thorough evaluation of benchmark methods on the proposed dataset and reveal their limitations on text manipulation detection.
3. We propose an extensible framework that takes advantage of the complementarity of various transformed domains. A plug-in-play Tampered-Authentic Contrastive Learning module further explicitly increases its discrimination ability. Experiments demonstrate notable performance improvement, especially on manual manipulations.

2. Related work

2.1. Datasets for image manipulation detection

Image manipulation techniques can be briefly categorized into copy-move, splicing and removal [14]. Numerous related datasets have significantly facilitated the progress of the research community, such as Columbia [15], COVER [6], CASIA [5], NIST16 [7], RTD [8], *etc.* However, these works mainly focus on tampering with semantic objects in natural images. Thereby, they hardly reflect the individual features of text manipulations. Therefore, there is a need for datasets tailored for text manipulation detection. Besides, several IMD datasets have taken consideration of handcraft manipulations, which is still a gap in text manipulation datasets, as shown in Table 1. Inspired by this, our proposed dataset focuses on real-life text forgery.

2.2. Datasets for text manipulation detection

Most of the early works [16–19] are conducted on private datasets, while proper data is crucial for developing detection methods. In recent years, as document security has attracted growing attention, sufficient datasets are urgently needed. Competitions [20,21] organized by industries show broad interest, while access to their data is limited to the contest. In academia, most datasets are generated by synthetic approaches due to the high cost of text manipulation and annotations. Sidere et al. [22] perform text forgery on synthetic payslips with fixed layouts. Find it! [12] and Find it again! [13] propose datasets for detecting forged texts in receipts and additional textual transcriptions are provided as NLP analysis assistance. Recently, T-SROIE [2] and

Table 1
Statistic comparison between existing datasets and RTM. 'Manual Tamper', 'Automatic Tamper' and 'Authentic' are abbreviated as 'MT', 'AT' and 'Auth.'. Area indicates the average relative area of tampered regions per image. CM, SP, GN, CV, IP denote Copy-move, Splicing, Generation, Coverage and Inpainting, respectively.

Scene	Dataset	Year	Statistics				Tampering Methods					Properties	
			Total	MT	AT	Auth.	CM	SP	GN	CV	IP	Area	Irregular
Natural image	CASIAv2 [5]	2013	12,615	5,123	0	7,492	✓	✓				8.96%	✓
	COVERAGE [6]	2016	200	100	0	100	✓					11.17%	✓
	NIST16 [7]	2019	1,124	564	0	560	✓	✓			✓	7.23%	✓
	RTD [8]	2019	440	220	0	220	✓	✓			✓	5.72%	✓
Text image	Find it! [12]	2018	1,180	240	0	940	✓	✓	✓	✓	✓	–	
	T-SROIE [2]	2022	986	0	986	0			✓			0.95%	
	T-IC13 [10]	2022	462	0	378	84			✓			9.28%	
	DocTamper [11]	2023	170,000	0	170,000	0	✓	✓	✓			1.67%	
	Find it again! [13]	2023	988	163	0	825	✓	✓	✓	✓	✓	1.11%	
	RTM (ours)	–	9,000	6,000	0	3,000	✓	✓	✓	✓	✓	1.18%	✓

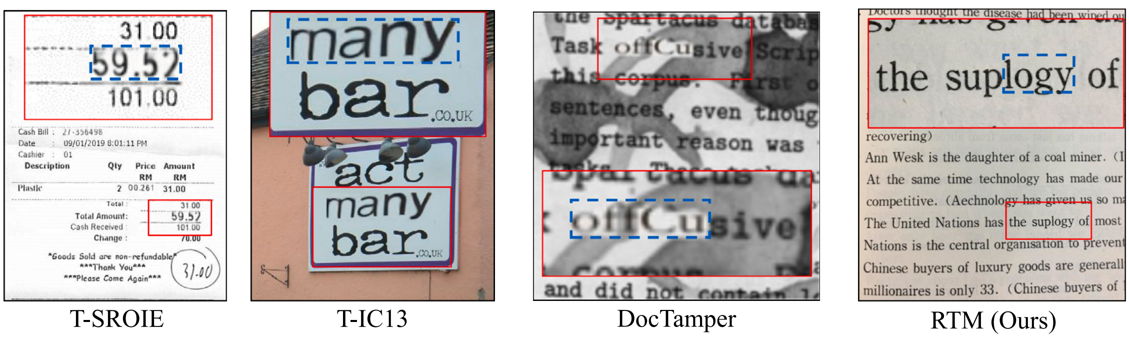


Fig. 2. Visualization comparison of tampered text generated by synthetic approaches in previous dataset [2,10,11] and manually tampered text in our RTM. The red box zooms in the regions that contain tampered text indicated by blue dashed boxes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

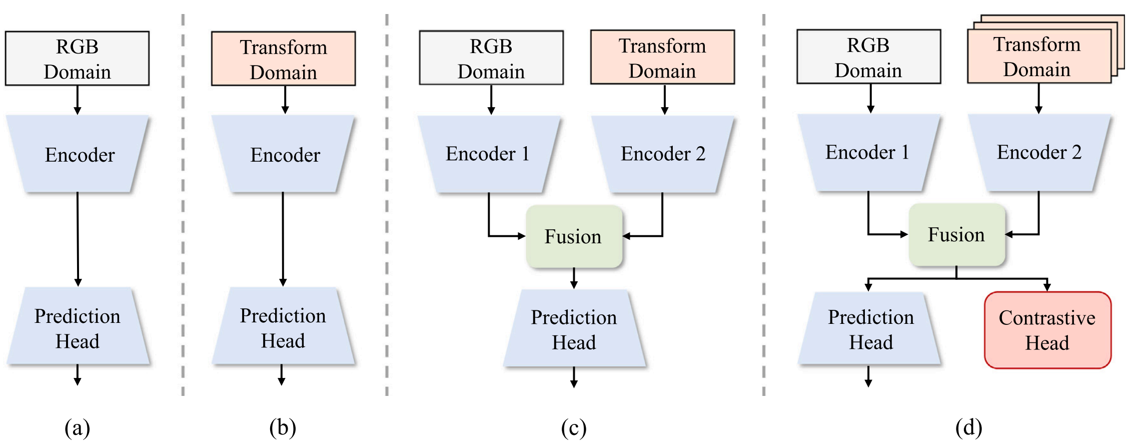


Fig. 3. Popular frameworks for text manipulation detection. (a) and (b) are single-stream architectures that trace the manipulation cues from RGB and transformed domains, respectively. (c) is the dual-stream architecture that combines the information from both domains. (d) is our asymmetric dual-stream framework extendable to multiple cues. Besides, a contrastive head is introduced during training.

T-IC13 [10] propose tampered text datasets through a text synthesis approach [23], effectively contributing to the community. Qu et al. [11] design a new approach to generate more realistic tampered documents. However, generative methods would inevitably introduce bias. As in Fig. 2, The tampered text in T-SROIE and T-IC13 exhibit blurry effects caused by GAN-based synthesis. Unlike randomly choosing a target area, human operators can place the forged text in a more aligned position. Besides, the manipulations based on bounding boxes restrict their diversity, especially for tampering with irregular shapes. Such limitations make them insufficient to reflect the characteristics of real-world text manipulation properly. Therefore, we construct RTM dataset with real-mode forgery performed by skilled fraudsters, which is carefully designed to exhibit the features and challenges as illustrated in Section 1.

2.3. Text manipulation detection methods

Unlike traditional OCR tasks [24–26] that aim to understand the contents of texts, the goal of text manipulation detection is to determine whether the texts in images are modified. Early works leverage particular handcraft features to detect the distortions introduced by the tampering operations. Some [16,17] convert this task to printer source identification by recognizing external prints as forged texts. Others investigate the distortions in fonts [27], text line orientation [28], geometry [18], image quality [29], DCT coefficient [30] and local texture pattern [31] to detect tampered text. These methods have high interpretability, but their generalization is limited when facing more concealed cases.

With the advances in deep learning and its effectiveness in various vision tasks, modern arts use CNNs or Transformers to extract the forgery features. As shown in Fig. 3, previous methods can be briefly categorized as single-stream and dual-stream architecture. The former takes the original image as the input [32,33] or preprocesses the image into a transformed domain to enhance the manipulation trace from a certain perspective before feeding it into the encoder. Liang et al. [33] leverage various attention modules to capture discriminative features of tampered texts of diverse scales. Tornés et al. [34] further boost the performance of detecting forged receipts by introducing language models.

Inspired by image manipulation detection methods designed for nature images [35–37], the most recent methods employ a dual-stream encoder to extract the manipulation cues from RGB and transformed domain respectively, combining the complementary information from both views. Xu et al. [38] utilize residual filters in the second stream to capture the manipulation traces in pixel correlation. Some integrate frequency information using discrete cosine transform [2] and high-pass filter [10]. STFL-Net [19] detects tampered text in screenshots by aggregating RGB and OCR information. Qu et al. [11] propose using JPEG compression trace to enrich forgery cues along with a curriculum learning procedure. Recently, some works [39] argue that the information from transformed domains is not always supportive. In the document scenario, we think the additional cues are still helpful under a more flexible approach. Therefore, aiming to tackle the nature of concealment and diversity of text manipulation, we explore the complementarity of RGB and other domains using an asymmetric dual-stream architecture. CA Hub and GCNF are designed to adaptively select and fuse informative traces and seek subtle signs of tampering. Moreover, We introduce a Tampered-Authentic Contrastive Learning module to further increase the distinction of feature representations explicitly.

3. RTM dataset

This section describes the proposed RTM from three perspectives: dataset construction, statistical analysis, and evaluation protocols.

3.1. Dataset construction

The proposed RTM dataset is built through the following three steps. Firstly, we first collected a large number of text images from different sources by considering the diversity of scenarios. Secondly, various types of text manipulations were performed on the collected images. Thirdly, several transformations were employed to further simulate the distortions in actual applications.

Text Image Collection. We collected text images from three sources to create diversity: (1) common text images from real applications provided by a famous e-commerce platform after data desensitization, which included certifications, contracts, invoices, screenshots, etc.; (2) text images photographed by 15 volunteers using smartphones and digital cameras from daily life, including books, commodity packages, papers, signboards, etc; (3) documents selected from the open-source datasets, particularly, receipts from SROIE [40], scanned documents from FUNSD [41] and TNCR [42]. The diverse sources provide various text images with different backgrounds, layouts, languages and styles that challenge the robustness of detectors. Fig. 4 shows some typical types of collected text images, such as charts, receipts, certificates, *etc.* The documents comprise both photographed and scanned images. For the data desensitization, we simply crop out the sensitive regions or remove the whole image to avoid confusion with tampering.

Text Manipulation. Aiming to simulate real-world text tampering, we perform manual manipulations on the collected text images. Before that, we have summarized five of the most common techniques based on the observation of practical applications as illustrated in Fig. 5.

(1) Copy-move: The operator copied a text region (a randomly selected character, word or sentence) and pasted it to the same image without destroying the layout. As shown in Fig. 5(a), the text region indicated by the yellow line is copied and pasted to a carefully selected position. Since the orientation, font, size and color of the copied text region are consistent with the text around it, the overall structure of the image is well-kept, increasing the difficulty of text manipulation detection.

(2) Splicing: A text region from a source image was copied and pasted to a target image. It is a common text manipulation in the real world. As shown in Fig. 5(b), a handwritten signature was copied from a source document and pasted to a target document for forging the certification.

(3) Insertion: The operator inserted a new text, *e.g.* a word or sentence, on a image. As shown in Fig. 5(c), the added texts have similar colors and fonts with the text around.

(4) Inpainting: The operator removed a text region via the Content-Aware Fill tool of PS. In particular, as shown in Fig. 5(d), a text region was removed by filling these pixels based on the pattern of the reference background, *i.e.* the green area. To increase visual consistency, operators were asked to carefully select the reference region that excludes other text to avoid confusion.

(5) Coverage: The operator removed a text region by covering a highly similar background region on it, as shown in Fig. 5(e).

Moreover, combining these fundamental operations can create more complex tampering techniques. For instance, inpainting the original text from an image and then inserting new content can be an effective implementation of text replacement.

Manual manipulations were conducted by 25 professional operators with different editing preferences to get realistic falsifications. The operators edited the text images using Adobe® Photoshop (PS) empirically following the pre-defined manipulation operations above. They were asked to create diverse manipulations while simulating cases foreseeable in actual applications. Specifically, some manipulations maintain the document semantics, *e.g.* inserting the signature, forging the date as shown in Fig. 5(b) and (c), or changing the price on the receipt. Others are forged with random contexts. The core requirement is preserving high visual consistency within the tampered images.

Authentic samples are further introduced to validate the false positive rate of different approaches, as a high false alarm is a great concern that restricts the practicality of solutions.

Post-processing. Text images are often subjected to some processing in real-world applications, such as affine transformations and compression, leading to the distortion of the manipulation trace, further increasing the difficulty of manipulation detection. Therefore, we perform a series of post-processing operations on the dataset. In particular, we randomly resize the text image within a limited range of side lengths (512 to 2000) with a ratio randomly sampled from 0.5 to 2.0 to create distortion. At last, the text images are compressed into JPEG format with a quality factor randomly sampled from 0.75 to 1.0. These hyper-parameters are chosen via the observation of practical e-commerce scenarios following [20].

Annotation. As our tampered regions are in various shapes compared to existing datasets, we naturally adopt the format of pixel-level binary masks as precise annotations. The criteria for labeling is that the manipulated pixels are labeled. A straightforward approach is directly subtracting the tampered image and the original one. However, ambiguous noise might occur. Thus we propose a more suitable solution as shown in Fig. 6 by utilizing PS layers. Specifically, each operation is implemented on an extra layer, where only the manipulated pixels are non-transparent. Naturally, we binarize the union of non-transparent pixels from each layer L_i $(i = 1, \dots, n)$ to produce precise annotations:

$$M = B(L_1) \cup B(L_2) \cup \cdots \cup B(L_n), \tag{1}$$

where $B(*)$ denotes the binarization process, where transparent pixels are set to 0 while others are set to 1. Such a process can be performed

Fig. 4. Illustration of the collected documents.

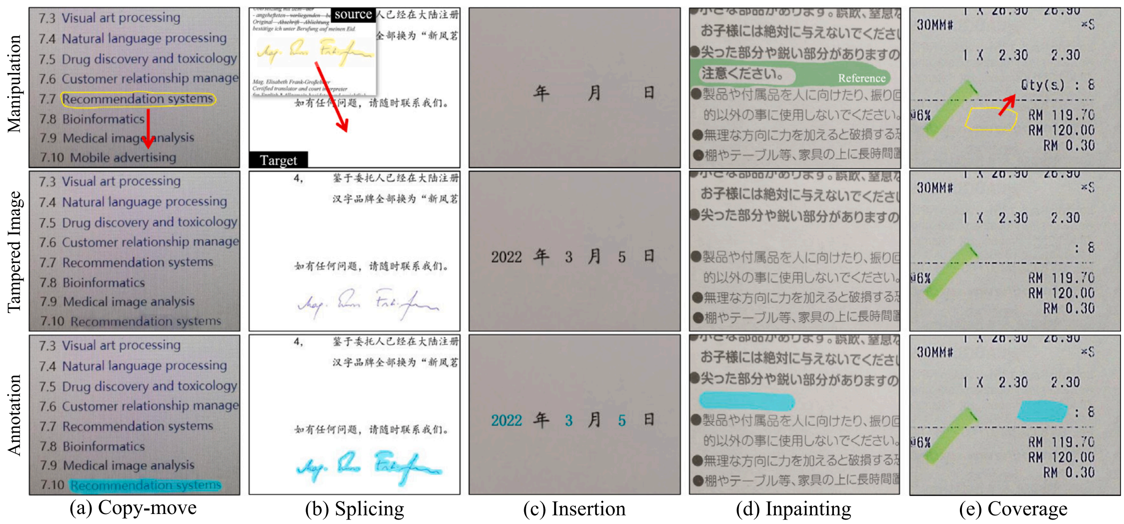


Fig. 5. Illustration of manual manipulation operations. The yellow lines in (a), (b) and (e) circle the copied regions, the green area in (d) is the reference for inpainting, and the cyan regions on the bottom row are annotations. Red arrows indicate the moving direction. All displayed samples are cropped from the original images and zoomed in for better visualization. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

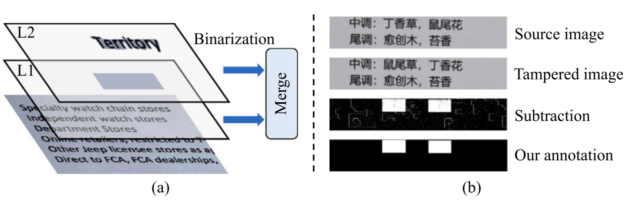


Fig. 6. (a) illustrates our annotating method using PS layers by aggregating non-transparent pixels. (b) compares the annotations generated by subtraction and ours.

in parallel with the above manipulation operations. Hence, consistent annotations are generated.

Finally, we check the manipulated images and annotation, filtering out the images with sensitive contents or of poor quality. As a result, RTM dataset contains 14,250 text images in total, including 6,000 manually manipulated text images and 3,000 authentic images. The dataset is further randomly divided into 5,803 training images and 3,197 testing images in the same proportion of each part.

3.2. Dataset analysis

Statisics. We split the dataset into 5,803 and 3,197 images for training and testing under the criteria of balanced distribution in each operation, as shown in Fig. 7(a). In the test set, the ratio of the tampered images and authentic images is around 1:2 since the forged documents are a minority in practice. Table 2 shows the number of images and regions of each tampering operation, where the regions are separated by connected component analysis. As most manipulations are performed by regions, we first mask the tampered areas with ground truth and count the number of tampered characters and words using PP-OCR [43]. In general, there are 7,502 forged words and 47,468 forged characters with the tampered regions. The side length of each sample is between 512 and 2000, and the aspect ratio is between 0.5 and 2.0. As shown in Fig. 7(b), the decentralized distribution represents high diversity, challenging detecting algorithm adaptive to various inputs. In addition, we maintain a wide range of the tampered region size to approximate the actual cases, while constraining the relative size of tampered regions below 50% to avoid confusion. Specifically, the relative area of tampered pixels is between 0.0055% to 41.4%, showing significant variation. As shown in Fig. 7(c), most tampered images are manipulated with less than 2% pixels. In particular, the smallest area of the forged instance is less than 64 pixels. Detecting small objects has been proved challenging in computer vision, such a problem is also a concern in TMD.

Analysis. We summarize the characteristics of RTM dataset into Diversity and Concealment.

Diversity. The diversity of RTM can be viewed in a variety of aspects. First of all, the collected pictures are from various sources, resulting in diverse languages and layouts. Secondly, we leverage several types of manipulation techniques, subjectively chosen by the operators with different editing preferences, creating diversiform forgery instances. Thirdly, the size and shape of tampered regions differ. The size of text in images already varies in documents according to different layouts. Moreover, the granularity of tampered regions has large variations. As shown in Fig. 8, the target can be characters, words, text lines, or even sentences. Another property that differentiates RTM from previous datasets is that we allow tampered regions with arbitrary shapes, including both regular shapes like rectangles with diverse aspect ratios

Table 2
Statistics of each tampering operation.

Operation	Copy-move	Splicing	Generation	Coverage	Inpainting	Combination	All
Abbreviation	CM	SP	GN	CV	IP	CB	–
Images	2,977	615	354	1032	783	239	6,000
Regions	6,706	962	3,018	1,353	1,132	295	13,466

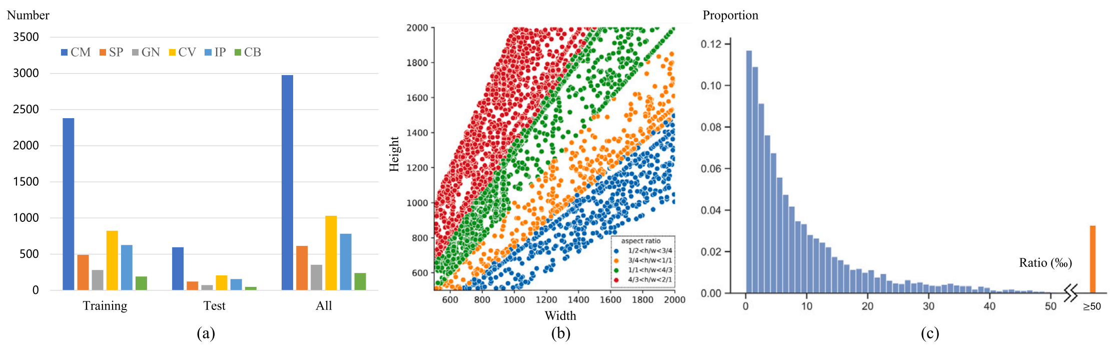


Fig. 7. Illustration of RTM statistics. (a) reports the data distribution in each subset; (b) is the distribution of image size and aspect ratio; (c) is the distribution of the relative area of tamper regions.

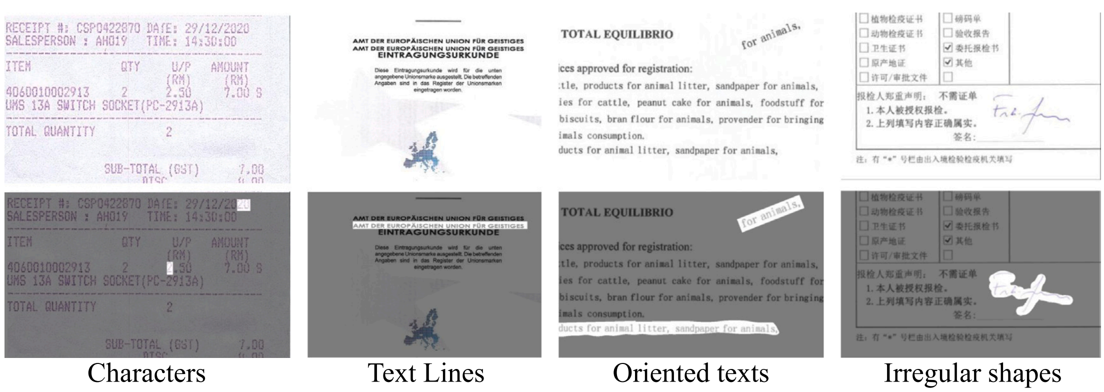


Fig. 8. Samples from RTM dataset, illustrating large scale and shape variations. The top row gives the tampered images and the bottom row visualizes the corresponding ground truth.

and irregular areas that fit the contours of texts, such as the handwritten signature in Fig. 8. Highly diverse data distribution contributes to the simulation of actual scenarios, challenging the generalization of solutions.

Concealment. Concealment is one of the major characteristics of text manipulation, which is reflected in various aspects. Firstly, the relative area of a forgery is small in most cases, leaving subtle manipulation. Secondly, the tampered text has a highly similar appearance to the surrounding content. In some cases, the color, font and orientation of the forged text are even the same as the surroundings, increasing the difficulty of distinguishing their differences. Thirdly, the tampered regions in most cases have weak contrast between their boundaries due to the similar and consistent background. Such difficulty is evident, especially on some screenshots and scanned documents that have passed through the denoising process because their background is purer. Fourthly, the distortion brought by post-processing further weakens the trace of manipulation. It is challenging to detect subtle artifacts with such little visual difference.

3.3. Evaluation protocols

To comprehensively validate the effectiveness of the text manipulation detection approaches, we design the evaluation protocols from the following aspects.

Pixel-level Metric. Following the widely used protocols in IMD methods, We leverage the intersection over union (IoU) to evaluate the localization performance:

$$IoU = \frac{TP}{TP + FP + FN}. \tag{2}$$

In addition, we use pixel-level precision (P), recall (R) and F1-measure (F1) to give a detailed analysis of the localization performance:

$$P = \frac{TP}{TP + FP}, \tag{3}$$
$$R = \frac{TP}{TP + FN}, \tag{4}$$
$$F1 = \frac{2 \times P \times R}{P + R}. \tag{5}$$

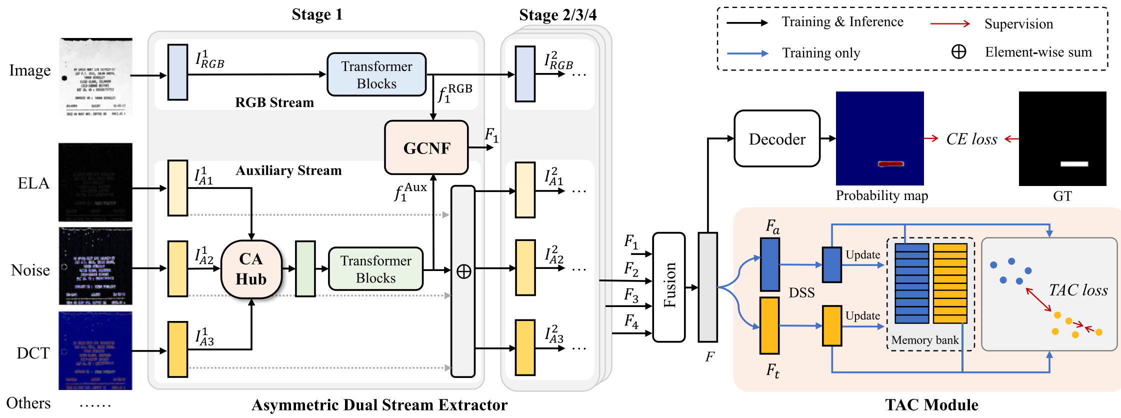


Fig. 9. Pipeline of our proposed baseline method. The meta structure of stage 2/3/4 is the same as stage 1. Down-sampling operations between stages are the same as MIT-B2. In each stage, CA Hub aggregates the informative cues from the transformed domains, GCNF adaptively injects auxiliary information into the features extracted by RGB Stream. TAC module explicitly enhances the latent representations to discriminate tampered regions.

Image-level Metric. As image-level performance is also concerned in practical applications, our dataset is constructed with authentic samples. We utilize F1-measure following previous works [35,36] to evaluate the classification performance.

4. Methodology

Aiming at the challenges above, we build an extensible baseline, namely, Asymmetric Stream Contrastive Transformer (ASC-Former).

4.1. Overview

The main pipeline of ASC-Former is illustrated in Fig. 9, which consists of a Dual Stream Extractor, GCNF for feature fusion, TAC Module, and a decoder to predict tampered regions. Given a suspicious text image $I \in \mathbb{R}^{H\times W\times 3}$, we apply various image transformations such as block-wise discrete cosine transform, SRM filter and error level analysis [44] to trace manipulation artifacts from different aspects. Dual Stream Extractor obtains hierarchical features from RGB and the transformed domains, where CA Hubs are designed to aggregate cues before each stage of the auxiliary stream. Subsequently, the Gated Cross Neighborhood-attention Fusion module fuses the features from different streams. Multi-scale features are then fused into F following SegFormer. Finally, we simply use a lightweight decoder to decode F and predict tampered text regions. During training, F is fed into TAC Module for contrastive learning.

4.2. Asymmetric dual stream extractor

The Asymmetric Dual Stream Extractor consists of an RGB Stream and an Auxiliary Stream. RGB Stream captures the semantic visual cues of inconsistent text style, while the Auxiliary Stream provides complementary cues from low-level patterns. We briefly select three representative modalities as follows:

(1) *Frequency domain* is widely used in image forensics as an additional cue. We leverage block-wise discrete cosine transform (DCT). It provides the cue from compression artifacts since manipulation operation might disrupt the block effect left by image compression. It also enhances the representation of text stroke since sharp texts are usually high-frequency parts of documents.

(2) *Steganalysis rich model (SRM)* [45] extracts the semantic-agnostic texture properties of a digital image introduced by the sensor of the camera. It models the relation of adjacent pixels with residual filters and reflects the high-frequency noise as described in [45].

(3) *Error level analysis (ELA)* [44] works by intentionally re-saving the image at a known error rate, and then computing the difference between the images. The tampered and authentic regions exhibit different sensitivities toward double compression. As compression usually discards high-frequency information, such a phenomenon is valid to texts.

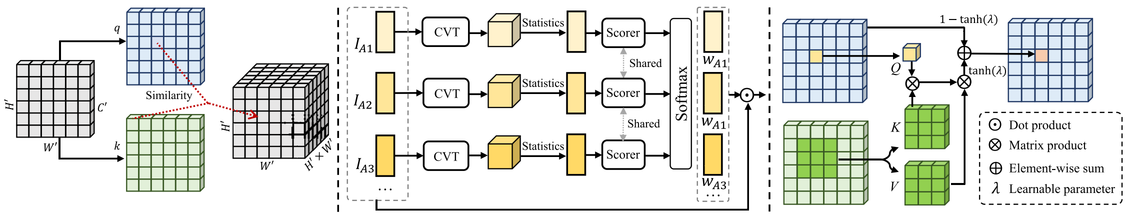
RGB Stream extracts hierarchical features f_i^{RGB} $(i = \{1,2,3,4\})$ from RGB space of input image $I \in \mathbb{R}^{H\times W\times 3}$. As Forgery is usually the opposite concept of authenticity, it is essential to modulate intra-relations. Hence, we adopt MIT-B2 as the backbone due to several advantages. Firstly, Transformers are naturally capable of modulating intra-relation for inconsistency perception. Secondly, the convolution-based positional encoding is friendly to input images of arbitrary resolutions. Thirdly, the hierarchical architecture suits the scale variation of tampered regions.

Auxiliary Stream take all other transformed domains $I_{An} \in \mathbb{R}^{H_n\times W_n\times C_n}$ as the inputs. The architecture of the Auxiliary Stream is modified from the RGB Stream by introducing a Consistency-aware Aggregation Hub (CA Hub) before the Transformer blocks of each stage to select and fuse helpful cues from transformed domains. Subsequently, the fused embedding is fed into Transformer blocks of each stage, producing f_i^{Aux} $(i = \{1,2,3,4\})$. After each stage, the embedding of each domain is updated by f_i^{Aux}.

Consistency-aware Aggregation Hub selects the representative information from transformed domains to aid the RGB stream. Unlike multi-modal information from different sensors, the transformed domains are all derived from the original RGB image and each represents a certain perspective of it. We hypothesize that an effective domain can amplify the inconsistencies between tampered and original regions. Therefore, we describe the variation in a transformed domain by Consistency Volume Transformation (CVT), which is shown in Fig. 10(a). The latent representation of one transformed domain is first projected into the metric space as q and k. Then We calculate the cosine similarity of each vector on q and k, forming a 4-D consistency volume $V \in \mathbb{R}^{H'\times W'\times H'\times W'}$ that maintains spatial information:

$$V_{i1,j1,i2,j2} = \frac{q_{i1,j1}\cdot k_{i2,j2}}{\parallel q_{i1,j1}\parallel \cdot \parallel k_{i2,j2}\parallel}, \tag{6}$$

where $i1, i2 \in \{1,2,\ldots,H'\}$ and $j1, j2 \in \{1,2,\ldots,W'\}$ are the vertical and horizontal coordinates of vectors on q and k.



(a) Consistency Volume Transformation (CVT) (b) Consistency-aware Aggregation Hub (c) Gated Cross Neighborhood-attention Fusion

Fig. 10. Details of the modules in our pipeline. (a) Consistency Volume Transformation, where the last two dimensions are merged as the depth for better visualization; (b) Consistency-aware Aggregation Hub; (c) Gated Cross Neighborhood-attention Fusion, where FFN and normalization layer are omitted for better visualization.

Since we expect the transformed domain to provide informative and complementary cues, we hypothesize that the higher difference of vectors indicates the sensitivity of a transformed domain. Therefore, we obtain the statistical information of each position, including mean, variance, maximum and minimum value. They are further fed into a shared Scorer to predict the importance of each modality. Softmax is leveraged to normalize the weight of each modality.

4.3. Gated neighborhood cross-attention fusion

As two streams contain manipulation cues from different views, we fuse the deep features through cross-attention mechanism [46] for flexibility. Inspired by Neighborhood Attention (NSA) [47], we propose a Gated Cross Neighborhood-attention Fusion module (GCNF), as shown in Fig. 9. the features from the same level f_i^{RGB} and f_i^{Aux} are first enhanced by NSA to model intra-adjacent relation. Then, the enhanced features $\hat{f}_i^{RGB}$ and $\hat{f}_i^{Aux}$ are fed into CNA layer for fusion. In practice, we utilize $\hat{f}_i^{RGB}$ as the query, and $\hat{f}_i^{Aux}$ as the key and value, mathematically:

$$A_p^k = \begin{bmatrix} Q_p K_{n_1(p)}^T + B_{(p,n_1(p))} \\ Q_p K_{n_2(p)}^T + B_{(p,n_2(p))} \\ \vdots \\ Q_p K_{n_k(p)}^T + B_{(p,n_k(p))} \end{bmatrix}, \tag{7}$$

where Q_p and $K_{n_k(p)}$ are the projections of the pth vector of $\hat{f}_i^{RGB}$ and the kth neighbor of the pth vector of $\hat{f}_i^{Aux}$, respectively. $B(p,j)$ indicates the relative positional biases. Correspondingly, $\hat{f}_i^{Aux}$ are projected as the value:

$$V_p^k = \begin{bmatrix} V_{n_1(p)}^T & V_{n_2(p)^T} & \cdots & V_{n_k(p)^T} \end{bmatrix}^T. \tag{8}$$

Thereby, the cross-neighborhood attention is formulated as:

$$CNA_k(Q,K,V) = softmax(\frac{A_p^k}{\sqrt{d}})V_p^k. \tag{9}$$

A learnable parameter λ is introduced to control the fusion by gating mechanism. Hence, the fusion process of level i is formulated as:

$$f_i = (1-tanh(\lambda)) \cdot \hat{f}_i^{RGB} + tanh(\lambda) \cdot FFN(CNA(\hat{f}_i^{RGB}, \hat{f}_i^{Aux}, \hat{f}_i^{Aux})), i \in \{2,3,4\}. \tag{10}$$

4.4. Tampered-authentic contrastive learning

To further address the problem of concealment, we introduce a Tampered-Authentic Contrastive Learning Module (TAC Module) to increase the distinctiveness of the fused features explicitly. Inspired by [48] which extends InfoNCE [49] to receive multiple positive pairs, we propose a Tampered-Authentic Contrastive Loss (TAC Loss) to achieve our motivation. In short, the patch features belonging to the same class in a mini-batch are positive samples, while others are negative samples. However, there still remain two major difficulties, *i.e.* extreme data imbalance and high computational cost. As recent studies [50] demonstrate the negative impact of pooling when capturing subtle cues, we propose to obtain the positive and negative pairs via decoupled sampling. We also design a memory bank to provide more valid sample pairs.

Algorithm 1 Decoupled Sampling Strategy (DSS)

```
Input: Mini-batch of fused features F; Ground truth GT;
Output: The sampled feature vector for tampered regions S_t and authentic regions S_a.
Constant values: P_min, P_max.
Static method: RANDOM_SELECT(L, N) randomly selects N elements from L.
function DSS(F, GT)
    t_idx = (GT == 1)
    a_idx = (GT == 0)
    N_t = len(t_idx)
    N_a = len(a_idx)
    if min(N_t, N_a) < P_min then
        return
    else
        if N_t < P_max then
            t_idx = RANDOM_SELECT(t_idx, N_t)
            a_idx = RANDOM_SELECT(a_idx, 2 × P_max − N_t)
        else if N_a < P_max then
            t_idx = RANDOM_SELECT(t_idx, 2 × P_max − N_a)
            a_idx = RANDOM_SELECT(a_idx, N_a)
        else
            t_idx = RANDOM_SELECT(t_idx, P_max)
            a_idx = RANDOM_SELECT(a_idx, P_max)
        end if
        S_t = F[t_idx]
        S_a = F[a_idx]
        return S_t, S_a
    end if
end function
```

Firstly, as illustrated in Algorithm 1, we first pre-define the minimal and maximal number of sampled points P_{min} and P_{max}, the former controls when TAC Loss is applied. At the same time, the latter alleviates the computational cost. Given feature map F extracted from a mini-batch of images, we separate the feature vectors representing authentic and tampered regions according to the corresponding ground truth.

Because a vector must have both positive and negative pairs, P_{min} is larger than 2 in practice.

Algorithm 2 Memory Bank Updating Strategy of One Class

Input: Index of sampled features s_idx, Index of unsampled features u_idx, Features of tampered/authentic regions $\overline{F} \in (F_t, F_a)$, Memory bank M;
Output: The updated memory bank M.
Constant values: Maximum Number of stored vectors for each step N_{max}
Static method: $\textsc{Random\_Select}(L, N)$ randomly selects N elements from L.
function $\textsc{Update\_Memory}(\overline{F}, s\_idx, u\_idx, M)$
  $del(M[0])$
  $N_u = len(u\_idx)$
  if $N_u > N_{max}$ **then**
    $idx = \textsc{Random\_Select}(u\_idx, N_{max})$
  else
    if $N_u > 0$ **then**
      $idx = \textsc{Random\_Select}(s\_idx, (N_{max} - N_u))$
      $idx.extend(u\_idx)$
    else
      $idx = \textsc{Random\_Select}(s\_idx, (N_{max} - N_u))$
    end if
  end if
  $M.append(\overline{F}[idx])$
  return M
end function

Due to the rarity and size variation of tampered text regions, the number of samples in a mini-batch might be insufficient to support contrastive learning. Thus, we design a memory bank to save and provide valid and diverse sample pairs for each batch. The memory bank is updated every iteration to maintain its effectiveness following Algorithm 2. We prioritize the currently unsampled vectors to increase the diversity.

We then obtain the sets of authentic representation vectors S_a and tampered text S_t by merging the memorized and current samples. The vectors in each set are positive pairs and vectors from different sets are negative pairs. Thereby, TAC Loss is formulated as:

$$\mathcal{L}_{\text{TAC}} = \sum_{i \in I} \frac{-1}{N_p} \sum_{p \in P(i)} log \frac{exp(x_i \cdot x_p/\tau)}{\sum_{s \in P(i)} exp(x_i \cdot x_s/\tau) + \sum_{t \in N(i)} exp(x_i \cdot x_t/\tau)}, \tag{11}$$

where N_p is the number of positive pairs for the vector x_i, $P(i)$ and $N(i)$ are the positive and negative sets for x_i. Finally, the full optimization objective is the combination of the segmentation loss and the contrastive learning loss:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{TAC}}, \tag{12}$$

where $\mathcal{L}_{\text{CE}}$ is the cross entropy loss. λ is set to 0.05 by default. Note that the TAC Module can be removed during inference, thus no extra cost is introduced to deployment.

5. Experiments

5.1. Mainstream baseline methods

Due to insufficient open-source text manipulation detection methods, we evaluate our datasets using publicly available semantic segmentation, image manipulation detection methods, and document forgery detection methods. Semantic segmentation methods include classic CNN-based methods, *i.e.* UperNet [51], DeepLabV3+ [52], HRNet-OCR [53] as well as the latest Transformer-based methods, *i.e.* Segformer [54], MaskFormer [55] and Mask2Former [56]. As for image manipulation detection, we compare to recent open-source RRU-Net [57], PCSS-Net [58], MVSS-Net++ [35] and CAT-Netv2 [36]. For document forgery detection networks, We choose Liang et al. [33] and DTD [11].

5.2. Implementation details

In the experiments, the semantic segmentation methods are trained using the MMSegmentation Toolbox[1] and the default parameters. For the image manipulation detection methods, we use the official code of RRU-Net,[2] PSCC-Net[3] and CAT-Net v2.[4] MVSS-Net++[5] and DTD[6] only release their models, we reproduce the training procedure with the consistent settings. The code of Liang et al. [33] is reproduced as the official is unreleased. We keep simple data augmentations including random flip and random crop for fairness. CLTD [11] strategy is used in the training pipeline of DTD [11]. All the evaluated methods are trained using the training set of RTM. After obtaining a binary segmentation mask, image-level classification is determined by whether ‘tampered’ pixels occur on the mask. We train the models for 80k iterations with a batch size of 12, and other hyperparameters are set to default for each benchmark method. AdamW is used as the optimizer for our method, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay of 0.01. The learning rate is initialized as 6×10^{-5} and adjusted by “poly” learning rate policy.

5.3. Benchmark results

5.3.1. Quantitative results

We report the performance in Table 3. we validate using the IoU metric on each operation as well as the entire test set. We also report pixel-level precision, recall, F1-measure as well as image-level F1-measure on the entire test set to give a detailed analysis in Table 4. In general, semantic segmentation approaches are more likely to miss tampered instances, while forgery detection methods struggle with high false alarms. Surprisingly, we find image manipulation detection and document forgery detection methods do not dominate this task. SegFormer achieves the best overall localization performance among the existing methods. Transformer-based methods have higher recall than CNN-based methods, revealing the importance of relation modeling provided attention mechanism in this task. To satisfy the input of various resolutions, PSCC-Net resizes the deep features to a fixed size to avoid the excessive computational cost. However, due to the various sizes of images and tampered regions, this fixed scheme is less appropriate, especially for small text, resulting in unsatisfactory performance. Our method achieves the highest IoU of 19.71% and pixel-level F1-score of 32.93%, surpassing SegFormer by 5.17% in F1-score. We outperform the second-best by 3.91% in IoU on manual manipulations. Particularly, ASC-Former shows a significant improvement of 11.99% in pixel-level precision compared to the second-best. Due to the high pixel-level precision, the performance drop is much less when considering the authentic samples. The asymmetric architecture effectively integrates the cues from multi-view, contributing to an increase of 3.43% in pixel-level recall compared to SegFormer (see Table 4).

Considering the tampering techniques, copy-move, generation and coverage are more difficult to detect in general. Semantic segmentation methods are good at localizing text removal forgery, while image

[1] https://github.com/open-mmlab/mmsegmentation
[2] https://github.com/yelusaleng/RRU-Net
[3] https://github.com/proteus1991/PSCC-Net
[4] https://github.com/mjkwon2021/CAT-Net
[5] https://github.com/dong03/MVSS-Net
[6] https://github.com/qcf-568/DocTamper

Table 3
The IoU performance (%) of the evaluated methods. The best is in bold.

Method	Publication	CM	SP	GN	CV	IP	CB	Tamper	All
UperNet [51]	ECCV 2018	6.67	11.09	4.66	11.53	9.14	10.99	8.92	8.26
DeepLabV3+ [52]	ECCV 2018	7.32	12.75	2.57	9.16	10.74	13.69	9.26	8.56
HRNet-OCR [53]	TPAMI 2021	6.06	13.16	0.74	6.18	7.75	11.35	7.61	6.81
SegFormer [54]	NeurIPS 2021	14.01	25.51	15.77	13.22	27.60	17.53	17.66	15.72
MaskFormer [55]	NeurIPS 2021	18.27	22.11	16.95	11.11	17.28	18.58	17.23	13.72
Mask2Former [56]	CVPR 2022	15.43	20.37	12.96	14.49	23.31	16.59	17.18	12.35
RRU-Net [57]	CVPRW 2019	2.39	6.20	3.39	3.54	7.91	4.07	4.08	3.73
PSCC-Net [58]	TCSVT 2022	4.52	7.49	1.31	4.73	6.43	3.34	4.93	3.31
MVSS-Net++ [35]	TPAMI 2023	8.75	12.06	4.65	7.18	5.18	12.24	8.43	5.11
CAT-Net v2 [36]	IJCV 2022	11.7	17.28	6.62	10.97	15.48	8.09	12.64	11.30
Liang et al. [33]	TrustCom 2022	6.16	15.55	3.10	8.30	10.52	12.22	8.39	4.63
DTD [11]	CVPR 2023	7.65	12.93	6.26	8.79	10.91	6.76	8.98	6.51
Ours		**18.57**	**32.79**	**18.89**	**16.06**	**27.63**	**19.35**	**21.57**	**19.71**

Table 4
The pixel-level precision, recall, F1-measure and image-level F1-measure (%) of the evaluated methods on the entire test set of RTM. The best is in bold.

Method	Publication	P	R	F1	Image F1
UperNet [51]	ECCV 2018	32.49	9.98	15.27	49.11
DeepLabV3+ [52]	ECCV 2018	32.24	10.43	15.76	52.87
HRNet-OCR [53]	TPAMI 2021	24.15	8.67	12.76	47.00
SegFormer [54]	NeurIPS 2021	38.45	21.01	27.17	61.51
MaskFormer [55]	NeurIPS 2021	26.00	22.50	24.13	**68.83**
Mask2Former [56]	CVPR 2022	19.10	**25.89**	21.99	64.02
RRU-Net [57]	CVPRW 2019	15.20	4.72	7.20	40.97
PSCC-Net [58]	TCSVT 2022	3.59	30.28	6.41	68.71
MVSS-Net++ [35]	TPAMI 2023	7.34	14.41	9.73	54.87
CAT-Net v2 [36]	IJCV 2022	30.18	15.30	20.31	54.82
Liang et al. [33]	TrustCom 2022	5.37	24.99	8.84	59.64
DTD [11]	CVPR 2023	11.94	12.52	12.22	53.76
Ours		**50.44**	24.44	**32.93**	63.31

Table 5
Ablation study on each module.

Setting			IoU		P	R	F1	Image F1
Aux	GCNF	TAC	Tamper	All	All	All	All	All
			17.66	15.72	38.45	21.01	27.17	61.51
		✓	17.85	16.66	47.75	20.37	28.56	**64.77**
✓			18.57	16.31	35.16	23.31	28.04	60.39
✓	✓		20.56	18.93	46.42	24.22	31.84	61.04
✓	✓	✓	**21.57**	**19.77**	**50.44**	**24.44**	**32.93**	63.31

and document tampering detection methods are more likely to detect forgery within text regions as they tend to focus on high-frequency parts of the image, which is also reflected in the visualization results. Transformer-based models show higher performance across all manipulations, demonstrating the core idea of tracing the inner inconsistency of images is effective for all tempering techniques. The proposed ASC-Former combines the above strengths, leading to the best performance on every manipulation.

5.3.2. Visualization results

We visualize some typical cases of high-performing methods in Fig. 11. With the assistance of additional cues, IMD methods improve the discovery ability at the cost of higher false alarms. In general, our method is more likely to discover tampered regions while maintaining a rather low false positive rate.

Copy-move is challenging since the content is from the same image, which has the same visual appearance as the source text. In the first row, three tampered texts with different granularity are copied and pasted on the screen. Most methods fail to detect any of them. Our method manages to discover two forged instances. The second row is a challenging case since the scanned receipt has a purer background. Two texts with different fonts and orientations are copied. Forgery detection networks detect part of the long sentence and MaskFormer segments the whole sentence. ASC-Former successfully finds all the instances as it is contrapuntally designed to increase the discrimination power via TAC. The third and fourth rows show the cases of splicing. As splicing introduces external regions that might deviate from the target image, it is easier than copy-move forgery. Therefore, we select two common cases in practical applications, especially in certification forgery. Besides the common manipulation of regular shapes like text lines, ASC-Former also exhibits decent performance on those of irregular shapes. Both the signature and the seal are detected. Our method predicts a better shape close to the tampered region as we explicitly enhance the feature representation using TAC by the sampling mechanism to fit higher resolutions. In the fifth row, the small inserted word is localized by ASC-Former, while competitors fail to detect it or have intolerable false alarms. The last three rows are cases of text removal. We select images of different backgrounds, including a dark blue screenshot, a photographed noisy gray receipt and a scanned white report. The first two remove texts by covering them using background from the original images. Only our method successfully localizes the tampered regions on the screenshot. For the photographed receipt of noisy background, existing methods are capable of detecting the covered region more or less except MVSSNet++, which is disturbed by the text above. A similar result is shown in the last row of inpainting forgery, where our method predicts a more accurate localization than others. The asymmetric architecture benefits from multiple transformed domains to improve the generalization against diverse manipulations. In addition, we find that semantic segmentation methods are good at localizing traces left by text removal.

5.4. Ablation study

We first conduct ablation studies and report the result in Table 5 to justify the effectiveness of each module. The information from the auxiliary stream effectively boosts the performance on manual manipulations, earning an overall improvement of 0.91% on IoU, mainly due to the 2.30% gain in recall, proving the validity of transformed domains to perceive manipulation traces from inconsistency. However, directly aggregating different cues by concatenation causes a drop in precision. The proposed GCNF and TAC further improve the performance from the perspective of feature fusion and discrimination. Both of them significantly contribute to the increment of precision. As a result, ASC-Former surpasses the baseline by 5.76%. Next, we give a deeper analysis of each component.

Complementarity of transformed domains. To explore the complementarity of different domains and the function of CA Hub, we build the variants from the baseline and report the results in Table 6. The selected three domains can be regarded as the representative

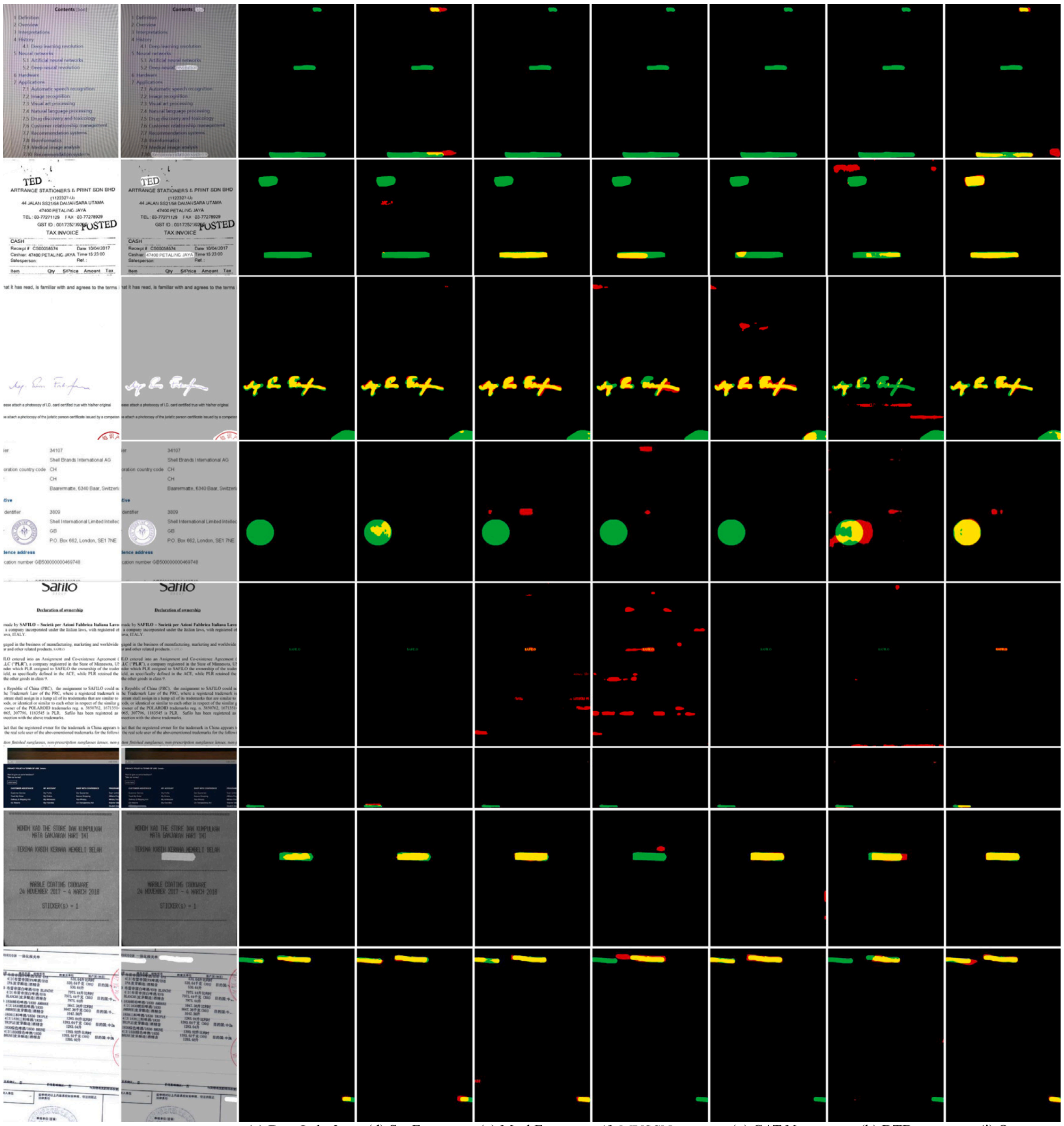

(a) Image (b) GT (c) DeepLabv3+ (d) SegFormer (e) MaskFormer (f) MVSSNet++ (g) CAT-Net (h) DTD (i) Ours

Fig. 11. Qualitative result of the evaluated methods. (a) is the test image, (b) is the corresponding ground truth. In (c)–(i), TP, FP, FN are in yellow, red and green, respectively. The ground truth is the union of TP and FN. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 6
Ablation study on the transformed domains and the CA Hub.

Setting			IoU		P	R	F1	Image F1
DCT	SRM	ELA	Tamper	All	All	All	All	All
			17.66	15.72	38.45	21.01	27.17	61.51
✓			18.38	16.64	50.08	19.95	28.53	62.58
	✓		19.58	17.48	44.68	22.30	29.75	61.17
		✓	20.26	18.21	44.31	23.62	30.81	59.43
✓	✓		20.04	18.02	40.59	24.48	30.54	62.82
	✓	✓	19.41	17.23	44.26	22.01	29.40	59.95
✓		✓	20.46	18.80	47.49	23.74	31.66	**63.10**
✓	✓	✓	**20.56**	**18.93**	**46.42**	**24.22**	**31.84**	61.04
Image as Aux			16.92	15.13	38.4	19.98	26.29	61.24
w/o CA			20.36	18.31	44.34	23.77	30.95	62.13

approaches of mainstream transformations. DCT describes the image from the view of frequency, SRM directly extracts the high-frequency patterns from the RGB domain with residual filters, and ELA reveals the manipulations through the changes in the image by compression. Compared to the baseline, each transformed domain can provide a performance gain. We also conduct an experiment of directly feeding the original image to the auxiliary stream (Image as Aux) to verify the contribution of asymmetric information rather than the increase of parameters. All the transformed domains achieve better performance on the manual manipulations. Moreover, the combinations of them achieve higher results. To verify the effectiveness of the CA Hub, we replace it by directly averaging the embeddings from different views, causing a performance drop in the recall of 0.45%, demonstrating the CA Hub is capable of selecting and aggregating the sensitive cues of tampering traces.

Fusion strategy. It is crucial to efficiently fuse the features from RGB and auxiliary domain, which determine whether the complementary information of manipulation cues from different is well integrated. In Table 7, we compare other fusion strategies including commonly used concatenation, addition, and multiplication as well as Transformer-based methods. We leverage the Efficient Attention [54] as

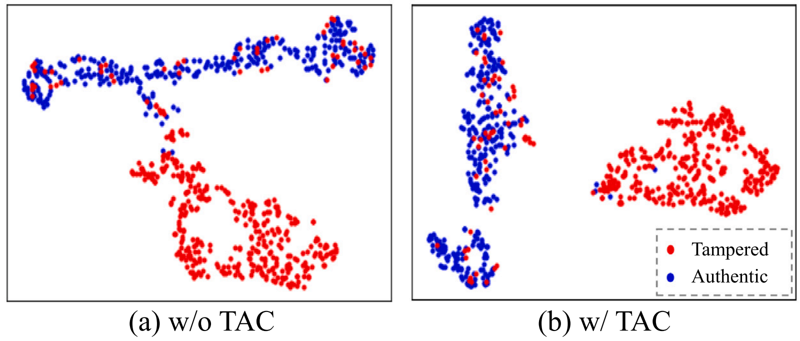


Fig. 12. t-SNE visualization of the deep features (a) without TAC, (b) with TAC.

Table 7
Ablation study on fusion strategies.

Fusion Strategy	IoU	Pixel-F1	Image-F1
Concat	16.31	28.04	60.39
Add	16.22	27.91	60.86
Multiply	16.99	29.05	62.06
Efficient Attention	16.46	28.27	61.22
GCNF	**18.93**	**31.84**	61.04
GCNF w/o gate	16.49	28.31	**63.01**
GCNF (Aux-query)	16.24	27.94	**63.01**
GCNF w/o gate (Aux-query)	12.25	21.83	54.95

Table 8
Comparison on public T-SROIE dataset.

Method	Precision	Recall	F1
EAST [59]	89.61	91.91	90.75
ATRR [60]	92.49	94.71	93.59
Wang et al. [2]	96.07	97.55	96.80
DTD [11]	99.23	**99.30**	**99.27**
Ours	**99.64**	98.31	98.97

Table 9
Comparison on public T-IC13 dataset.

Method	Precision	Recall	F1
m-PSENet	84.95	83.91	84.43
m-ATRR	86.10	90.84	88.40
Wang et al. [10]	88.43	91.85	90.11
DTD [11]	**92.17**	89.34	**90.73**
Ours	87.21	**92.21**	89.64

the representative of global paradigms to compare with our local neighborhood paradigm. In general, Transformer-based fusion strategies achieve better performance than traditional methods as the weights of the two streams are dynamic. We further investigate some detailed variants of the GCNF. We remove the gate mechanism from GCNF or switch the query and key/value. Both settings suffer a drop in localization performance. The gate introduces more flexibility to the fusion block and alleviates the interference of irrelevant information. After applying features from the auxiliary stream as the query, the IoU performance drops 2.69%, which justifies our assumption that regards the transformed domains as supplementary cues.

The effectiveness of TAC. By introducing TAC, IoU on manual tampering is improved by 1.01%, contributing to an overall performance gain of 0.84%, as shown in Table 5. Since TAC is a plug-in-play module for representation enhancement, we also directly apply it to the baseline, achieving 0.94% and 1.39% performance gain on IoU and pixel-level F1, respectively. Thanks to the explicit constraint on the feature representations to enlarge the discrepancy between tampered and un-tampered regions, it is easier for the model to classify forgery and genuineness. Particularly, precision is increased by 4.02% and 9.30% respectively after using TAC. We visualize F extracted from randomly selected samples using t-SNE in Fig. 12. After applying TAC during training, deep features are more distinct, leading to more precise results on some hard cases.

5.5. Discussion

Tampering techniques. RTM dataset consists of various tampering techniques to simulate forged documents in the real world. From Table 3, we find that splicing and inpainting forgery are more likely to be detected. The spliced text is from another image and inpainting removes texts with generated backgrounds. Therefore, more inconsistent patterns might be introduced from external content. Copy-move and coverage create forgeries using regions from the same image. The texture within them obtains higher similarity with surroundings, which raises the difficulty of detection, reflected by lower detection performance on these two techniques. Generation forgery directly inserts texts, the fidelity relies on the skill of the fraudster to select the proper font and color, *etc.*. However, the tampered areas are fine-grained and hardly include background, requiring models to discover subtle traces, especially when the inserted characters are relatively few. In general, text removal is harder since there is a lack of cues from the attributes of texts.

Experiments on synthetic datasets. We train ASC-Former on the GAN-based tampered text datasets T-SROIE [2] and T-IC13 [10], and report the performance in Table 8 and Table 9, respectively. Since they are manipulated with a fixed shape of horizontal rectangles and annotated with bounding boxes, we convert the tampered text boxes into binary masks to generate our labels. During inference, we first binarize the predicted mask and obtain the maximum circumscribed boxes of each connected component. Then, we filter out the tiny boxes as noises and follow the official evaluation protocol.[7] On document dataset T-SROIE, we achieve competitive results with DTD and have higher precision. T-IC13 is constructed on scene text, where the complex background introduces interference. Whereas, our method only leverages the tampered regions as supervision while most competitors also use the information of bounding boxes of authentic text. ASC-Former has a higher recall rate but suffers from false alarms, resulting in a comparable overall result.

Generalization experiment. Due to the diversity of tampering techniques in the real world, we evaluate the generalization of the high-performance methods in Table 10. They are trained using the training set of RTM and tested on the test set of T-SROIE. As T-SROIE is forged by region-based text synthesis method, we can convert the labels into masks and report the IoU. Although the tampering techniques are different, the evaluated models exhibit the generalization from manual tampering to unseen synthetic forgery. Among them, ASC-Former

[7] https://github.com/wangyuxin87/Tampered-IC13/blob/main/script.py

Table 10
Generalization experiment on RTM and T-SROIE, we report the IoU performance (%).

Method	SegFormer	MaskFormer	CAT-Net v2	DTD	Ours
IoU	64.6	71.89	55.03	55.41	**72.06**

achieves the best IoU. The performance differences between these two datasets also prove the difficulty of detecting manually manipulated text, demonstrating the potential of our RTM dataset to promote the development of text manipulation detection.

Comparison on manual and automatic tamper. Since manual text tampering requires professional skills to ensure enough fraudulence, both the manipulation process and annotating are costly. However, manual text tampering counts for a large portion and is an important object of study. From the results on our manually tampered dataset and synthetic dataset, a performance gap is observed, indicating the difficulty brought by manual forgery. The human prior contributes to the concealment of manipulation with more reasonable selections of tampered regions and manipulation techniques. Therefore, we build RTM dataset reflecting this property in the real world and hope it can promote the development of this field.

6. Potential social impact

To prevent the malicious use of the dataset, data desensitization is performed on the collected images by cropping off the sensitive information before text tampering. The released dataset can only be used for non-commercial algorithm research and educational purposes.

7. Limitations

We discuss the limitations from two perspectives, *i.e.* dataset and method. On the one hand, although our dataset concerns a variety of text manipulations, there are many more types in the real world due to the rapid progress in image editing techniques. In the future, we will leverage more tampering techniques and editing tools to expand the data scale. In addition, document attributes have the potential to assist in forgery detection. Hence, we will enrich the OCR annotations in the succeeding work. On the other hand, although our method makes an exploration of this task and achieves considerable progress over the baseline, there is still a large room to improve its accuracy and efficiency. Our method achieves comparable image-level performance, which is also important in actual scenarios. Therefore, we will improve its reliability by alleviating image-level false positives. Improving the contrastive learning structure is another direction of our future work to improve training efficiency. Besides, our method only leverages visual cues of tampered text, while the semantic context is also an important attribute in text manipulation detection. As our future direction, we will push our effort to combine natural language processing techniques or large language models to improve the understanding of tampered text.

8. Conclusion

The presented Real Text Manipulation (RTM) dataset addresses a critical gap by providing a substantial, varied collection of manually tampered text images alongside unaltered ones, facilitating a more realistic evaluation ground for text forgery detection methods. Our proposed baseline solution, integrating a Consistency-aware Aggregation Hub, a Gated Cross Neighborhood-attention Fusion module, and a Tampered-Authentic Contrastive Learning module, showcases promising strides toward enhancing detection and localization performance in confronting text tampering. We hope that our work will encourage further development of tampered text detection and contribute to information security in real life. However, ASC-Former still struggles with extremely few visual tampering cues. In future work, we plan to expand the data scale and annotations to develop our dataset. We also intend to incorporate natural language processing techniques to further explore the effectiveness of other attributes in text manipulation detection.

CRediT authorship contribution statement

Dongliang Luo: Methodology. **Yuliang Liu:** Writing – review & editing, Methodology. **Rui Yang:** Investigation. **Xianjin Liu:** Resources. **Jishen Zeng:** Funding acquisition. **Yu Zhou:** Supervision. **Xiang Bai:** Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data/code in the abstract.

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References

[1] P. Roy, S. Bag, Detection of handwritten document forgery by analyzing writers' handwritings, in: Proceedings of the International Conference on Pattern Recognition and Machine Intelligence, 2019, pp. 596–605.
[2] Y. Wang, B. Zhang, H. Xie, Y. Zhang, Tampered text detection via RGB and frequency relationship modeling, Chin. J. Netw. Inf. Secur. 8 (3) (2022) 29.
[3] L. Zhao, C. Chen, J. Huang, Deep learning-based forgery attack on document images, IEEE Trans. Image Process. 30 (2021) 7964–7979.
[4] X. Hou, L. Shen, Z. Ming, G. Qiu, Deep generative image priors for semantic face manipulation, Pattern Recognit. 139 (2023) 109477.
[5] J. Dong, W. Wang, T. Tan, Casia image tampering detection evaluation database, in: 2013 IEEE China Summit and International Conference on Signal and Information Processing, IEEE, 2013, pp. 422–426.
[6] B. Wen, Y. Zhu, R. Subramanian, T.-T. Ng, X. Shen, S. Winkler, COVERAGE—A novel database for copy-move forgery detection, in: Proceedings of the IEEE International Conference on Image Processing, IEEE, 2016, pp. 161–165.
[7] H. Guan, M. Kozak, E. Robertson, Y. Lee, A.N. Yates, A. Delgado, D. Zhou, T. Kheyrkhah, J. Smith, J. Fiscus, MFC datasets: Large-scale benchmark datasets for media forensic challenge evaluation, in: Proceedings of the IEEE Winter Applications of Computer Vision Workshops, IEEE, 2019, pp. 63–72.
[8] P. Korus, J. Huang, Evaluation of random field models in multi-modal unsupervised tampering localization, in: IEEE International Workshop on Information Forensics and Security, WIFS, IEEE, 2016, pp. 1–6.
[9] A. Novozamsky, B. Mahdian, S. Saic, IMD2020: A large-scale annotated dataset tailored for detecting manipulated images, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops, 2020, pp. 71–80.
[10] Y. Wang, H. Xie, M. Xing, J. Wang, S. Zhu, Y. Zhang, Detecting tampered scene text in the wild, in: Proceedings of the European Conference on Computer Vision, Springer, 2022, pp. 215–232.
[11] C. Qu, C. Liu, Y. Liu, X. Chen, D. Peng, F. Guo, L. Jin, Towards robust tampered text detection in document image: New dataset and new solution, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 5937–5946.
[12] C. Artaud, N. Sidere, A. Doucet, J.-M. Ogier, V.P.D. Yooz, Find it! fraud detection contest report, in: International Conference on Pattern Recognition, ICPR, 2018, pp. 13–18.
[13] B.M. Tornés, T. Taburet, E. Boros, K. Rouis, A. Doucet, P. Gomez-Krämer, N. Sidere, V.P. d'Andecy, Receipt dataset for document forgery detection, in: International Conference on Document Analysis and Recognition, Springer, 2023, pp. 454–469.
[14] F.Z. Mehrjardi, A.M. Latif, M.S. Zarchi, R. Sheikhpour, A survey on deep learning-based image forgery detection, Pattern Recognit. 144 (2023) 109778.
[15] Y.-F. Hsu, S.-F. Chang, Detecting image splicing using geometry invariants and camera characteristics consistency, in: Proceedings of the IEEE International Conference on Multimedia and Expo, IEEE, 2006, pp. 549–552.

[16] C.H. Lampert, L. Mei, T.M. Breuel, Printing technique classification for document counterfeit detection, in: 2006 International Conference on Computational Intelligence and Security, Vol. 1, 2006, pp. 639–644.
[17] M. Bibi, A. Hamid, M. Moetesum, I. Siddiqi, Document forgery detection using printer source identification—A text-independent approach, in: Proceedings of the International Conference on Document Analysis and Recognition Workshops, Vol. 8, 2019, pp. 7–12.
[18] S. Shang, X. Kong, X. You, Document forgery detection using distortion mutation of geometric parameters in characters, J. Electron. Imaging 24 (2015) 023008.
[19] Z. Yu, B. Li, Y. Lin, J. Zeng, J. Zeng, Learning to locate the text forgery in smartphone screenshots, in: IEEE International Conference on Acoustics, Speech and Signal Processing, 2023.
[20] Tianchi Algorithm Context, Real-world image forgery localization challenge, 2022, [Online], https://tianchi.aliyun.com/competition/entrance/531945/information.
[21] D. Luo, Y. Zhou, R. Yang, Y. Liu, X. Liu, J. Zeng, E. Zhang, B. Yang, Z. Huang, L. Jin, et al., ICDAR 2023 competition on detecting tampered text in images, in: Proceedings of the International Conference on Document Analysis and Recognition, Springer, 2023, pp. 587–600.
[22] N. Sidere, F. Cruz, M. Coustaty, J. Ogier, A dataset for forgery detection and spotting in document images, in: Seventh International Conference on Emerging Security Technologies, EST, 2017, pp. 26–31.
[23] L. Wu, C. Zhang, J. Liu, J. Han, J. Liu, E. Ding, X. Bai, Editing text in the wild, in: Proceedings of the ACM International Conference on Multimedia, 2019, pp. 1500–1508.
[24] M. Liao, Z. Wan, C. Yao, K. Chen, X. Bai, Real-time scene text detection with differentiable binarization, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 34, No. 07, 2020, pp. 11474–11481.
[25] B. Shi, X. Bai, C. Yao, An end-to-end trainable neural network for image-based sequence recognition and its application to scene text recognition, IEEE Trans. Pattern Anal. Mach. Intell. 39 (11) (2016) 2298–2304.
[26] Y. Liu, L. Jin, S. Zhang, C. Luo, S. Zhang, Curved scene text detection via transverse and longitudinal sequence connection, Pattern Recognit. 90 (2019) 337–345.
[27] R. Bertrand, O.R. Terrades, P. Gomez-Krämer, P. Franco, J.-M. Ogier, A conditional random field model for font forgery detection, in: Proceedings of the International Conference on Document Analysis and Recognition, 2015, pp. 576–580.
[28] J. Van Beusekom, F. Shafait, T.M. Breuel, Text-line examination for document forgery detection, Int. J. Document Anal. Recognit. (IJDAR) 16 (2013) 189–207.
[29] S.-J. Ryu, H.-Y. Lee, I.-W. Cho, H.-K. Lee, Document forgery detection with SVM classifier and image quality measures, in: Advances in Multimedia Information Processing-PCM 2008: 9th Pacific Rim Conference on Multimedia, Tainan, Taiwan, December 9-13, 2008. Proceedings 9, Springer, 2008, pp. 486–495.
[30] Z. Lin, J. He, X. Tang, C.-K. Tang, Fast, automatic and fine-grained tampered JPEG image detection via DCT coefficient analysis, Pattern Recognit. 42 (11) (2009) 2492–2501.
[31] F. Cruz, N. Sidère, M. Coustaty, V.P. D'Andecy, J.-M. Ogier, Local binary patterns for document forgery detection, in: Proceedings of the International Conference on Document Analysis and Recognition, Vol. 01, 2017, pp. 1223–1228.
[32] H. Joren, O. Gupta, D. Raviv, Learning document graphs with attention for image manipulation detection, in: International Conference on Pattern Recognition and Artificial Intelligence, Springer, 2022, pp. 263–274.
[33] W. Liang, L. Dong, R. Wang, D. Yan, Y. Li, Robust document image forgery localization against image blending, in: 2022 IEEE International Conference on Trust, Security and Privacy in Computing and Communications, TrustCom, 2022, pp. 810–817.
[34] B.M. Tornés, E. Boros, A. Doucet, P. Gomez-Krämer, J.-M. Ogier, Detecting forged receipts with domain-specific ontology-based entities & relations, in: International Conference on Document Analysis and Recognition, 2023, pp. 184–199.
[35] C. Dong, X. Chen, R. Hu, J. Cao, X. Li, Mvss-net: Multi-view multi-scale supervised networks for image manipulation detection, IEEE Trans. Pattern Anal. Mach. Intell. 45 (3) (2022) 3539–3553.
[36] M. Kwon, S. Nam, I. Yu, H. Lee, C. Kim, Learning JPEG compression artifacts for image manipulation detection and localization, Int. J. Comput. Vis. 130 (8) (2022) 1875–1895.
[37] X. Lin, S. Wang, J. Deng, Y. Fu, X. Bai, X. Chen, X. Qu, W. Tang, Image manipulation detection by multiple tampering traces and edge artifact enhancement, Pattern Recognit. 133 (2023) 109026.
[38] W. Xu, J. Luo, C. Zhu, W. Lu, J. Zeng, S. Shi, C. Lin, Document images forgery localization using a two-stream network, Int. J. Intell. Syst. 37 (8) (2022) 5272–5289.
[39] F. Li, Z. Pei, X. Zhang, C. Qin, Image manipulation localization using multi-scale feature fusion and adaptive edge supervision, IEEE Trans. Multimed. (2022) 1–15.
[40] Z. Huang, K. Chen, J. He, X. Bai, D. Karatzas, S. Lu, C. Jawahar, Icdar2019 competition on scanned receipt ocr and information extraction, in: Proceedings of the International Conference on Document Analysis and Recognition, IEEE, 2019, pp. 1516–1520.
[41] G. Jaume, H.K. Ekenel, J.-P. Thiran, FUNSD: A dataset for form understanding in noisy scanned documents, in: Proceedings of the International Conference on Document Analysis and Recognition Workshops, vol. 2, IEEE, 2019, pp. 1–6.
[42] A. Abdallah, A. Berendeyev, I. Nuradin, D. Nurseitov, TNCR: Table net detection and classification dataset, Neurocomputing 473 (2022) 79–97.
[43] C. Li, W. Liu, R. Guo, X. Yin, K. Jiang, Y. Du, Y. Du, L. Zhu, B. Lai, X. Hu, et al., PP-OCRv3: More attempts for the improvement of ultra lightweight OCR system, 2022, arXiv preprint arXiv:2206.03001.
[44] N.B. Abd Warif, M.Y.I. Idris, A.W.A. Wahab, R. Salleh, An evaluation of error level analysis in image forensics, in: IEEE International Conference on System Engineering and Technology, 2015, pp. 23–28.
[45] P. Zhou, X. Han, V.I. Morariu, L.S. Davis, Learning rich features for image manipulation detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 1053–1061.
[46] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Proceedings of the Advances in Neural Information Processing System, 2017, pp. 5998–6008.
[47] A. Hassani, S. Walton, J. Li, S. Li, H. Shi, Neighborhood attention transformer, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 6185–6194.
[48] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, D. Krishnan, Supervised contrastive learning, in: Proceedings of the Advances in Neural Information Processing System, Vol. 33, 2020, pp. 18661–18673.
[49] T. Chen, S. Kornblith, M. Norouzi, G.E. Hinton, A simple framework for contrastive learning of visual representations, in: Proceedings of the International Conference on Machine Learning, 2020, pp. 1597–1607.
[50] M. Boroumand, M. Chen, J.J. Fridrich, Deep residual network for steganalysis of digital images, IEEE Trans. Inf. Forensics Secur. 14 (5) (2019) 1181–1193.
[51] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, J. Sun, Unified perceptual parsing for scene understanding, in: Proceedings of the European Conference on Computer Vision, 2018, pp. 418–434.
[52] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, H. Adam, Encoder-decoder with atrous separable convolution for semantic image segmentation, in: Proceedings of the European Conference on Computer Vision, 2018, pp. 833–851.
[53] J. Wang, K. Sun, T. Cheng, B. Jiang, C. Deng, Y. Zhao, D. Liu, Y. Mu, M. Tan, X. Wang, W. Liu, B. Xiao, Deep high-resolution representation learning for visual recognition, IEEE Trans. Pattern Anal. Mach. Intell. 43 (10) (2021) 3349–3364.
[54] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J.M. Alvarez, P. Luo, SegFormer: Simple and efficient design for semantic segmentation with transformers, Adv. Neural Inf. Process. Syst. 34 (2021) 12077–12090.
[55] B. Cheng, A. Schwing, A. Kirillov, Per-pixel classification is not all you need for semantic segmentation, in: Proceedings of the Advances in Neural Information Processing System, Vol. 34, 2021, pp. 17864–17875.
[56] B. Cheng, I. Misra, A.G. Schwing, A. Kirillov, R. Girdhar, Masked-attention mask transformer for universal image segmentation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 1290–1299.
[57] X. Bi, Y. Wei, B. Xiao, W. Li, RRU-net: The ringed residual U-net for image splicing forgery detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2019.
[58] X. Liu, Y. Liu, J. Chen, X. Liu, PSCC-net: Progressive spatio-channel correlation network for image manipulation detection and localization, IEEE Trans. Circuits Syst. Video Technol. 32 (11) (2022) 7505–7517.
[59] X. Zhou, C. Yao, H. Wen, Y. Wang, S. Zhou, W. He, J. Liang, East: An efficient and accurate scene text detector, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2017, pp. 5551–5560.
[60] X. Wang, Y. Jiang, Z. Luo, C. Liu, H. Choi, S. Kim, Arbitrary shape scene text detection with Adaptive Text Region representation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 6449–6458.

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